

PLANETARY NEBULA

Bug Nebula



CATALOG NUMBER
NGC 6302

DISTANCE FROM SUN
4,000 light-years

MAGNITUDE 7.1

SCORPIUS

First discovered in 1826 by Scottish astronomer James Dunlop, then rediscovered in the late 19th century by the great American astronomer E. E. Barnard, the Bug Nebula is one of the brightest planetary nebulae. The central star is thought to have an extremely high temperature, and its intense ultraviolet radiation lights up the surrounding stellar material. The star itself is not visible at optical wavelengths because it is hidden by a blanket of dust. It is believed that the central star ejected a ring of dark material about 10,000 years before it was as we see it now, but astronomers cannot explain why it was not destroyed long ago by the star's ultraviolet emissions. The composition of the surrounding material is also surprising because it contains carbonates, which usually form when carbon dioxide dissolves in liquid water. Although ice exists in the nebula, along with hydrocarbons and iron, there is no evidence of liquid water.

COLORFUL BUG

The Bug Nebula is the ejected outer layers of a dying star that was once about five times the mass of the Sun. Ultraviolet radiation from the intensely hot central star is making the cast-off material glow.